

Group name

Enter your details :

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

Exploratory Data Analysis (2 of 2)

 Reflect and discuss (A)

 1. We took a random sample from the 2000 U.S. Census. Here is part of the dataset:

US Census 2000

	State	zipcode	Family_Size	Annual_income
1	Florida	32716	8	200
2	Alabama	35236	5	800
3	Florida	32116	6	13500
4	Florida	33679	5	21000
5	Alabama	36374	4	21000
6	California	94565	1	23000

Who are the _____ described by this data?

- A. People with families in the year 2000.
- B. States.
- C. People living in the United States in the year 2000.

 2. What type of variable is _____?

- A. Categorical.
- B. Quantitative.

 3. What type of variable is _____?

- A. Categorical.
- B. Quantitative.

 4. What type of variable is _____?

- A. Quantitative.
- B. Categorical.

A Sample Dataset

Reflect and discuss (B)

- *Hospt*: The patient's hospital, represented by a code for each of the five hospitals (1, 2, 3, 5, and 6)
- *Treat*: The treatment received by the patient (Lithium, Imipramine, or Placebo)
- *Outcome*: Whether a recurrence occurred during the patient's treatment (Recurrence or No Recurrence)
- *Time*: Either the time in days until the first recurrence or, if a recurrence did not occur, the length in days of the patient's participation in the study
- *AcuteT*: The time in days that the patient was depressed prior to the study
- *Age*: The age of the patient in years when the patient entered the study
- *Gender*: The patient's gender (1 = Female; 2 = Male)

 5. Who are the _____ described by this data?

- A. 109 clinically depressed people.
- B. 19 million adults who experience depression each year.
- C. Hospitals.

 6. Which of the following variables is _____? (select all that apply)

- A. Treat.
- B. AcuteT.
- C. Outcome.

 7. Which of the following variables is _____? (select all that apply)

- A. Hospt.
- B. Time.
- C. Age.
- D. Gender.

Module 4 | One Categorical Variable (2 of 3)

Reflect and discuss (C)

 8. What is the _____ between the two bar charts (see corresponding reading)?

- A. There is no difference.
- B. The two bar charts represent the distributions of two different variables.
- C. The first bar chart represents the count of respondents that chose each category, while the second bar chart represents the percentage of respondents that chose each category.
- D. The two bar charts represent the distribution of "Body Image" obtained from two different samples.

Now that we have summarized the distribution of values in the Body Image variable, let's go back and interpret the results in the context of the questions that we posed:

 Reflect and discuss (D)

 9. What do the results _____ about how the students are divided across the three body image categories?

- A. Students are equally divided across the three categories.
- B. Students are not equally divided across the three categories.

 10. How do the vast majority of students (71.3%) feel about their _____?

- A. About right.
- B. Overweight.
- C. Underweight.

 11. How does the middle group of students (19.6%) feel about their _____?

- A. About right.
- B. Overweight.
- C. Underweight.

 12. What was the body perception that occurred the _____ often?

- A. About right.
- B. Overweight.
- C. Underweight.

 Reflect and discuss (E)

 13. Here is the table from the corresponding reading, use it to answer the question.

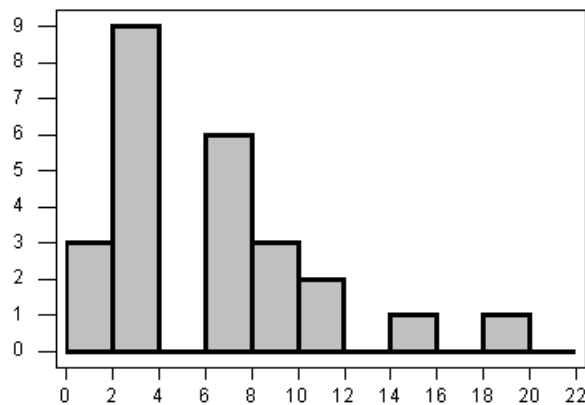
Score	Count
[40–50)	1
[50–60)	2
[60–70)	4
[70–80)	5
[80–90)	2
[90–100]	1

What percentage of students earned less than a grade of _____ on the exam?

- A. 9%.
- B. 15%.
- C. 47%.
- D. 53%.
- E. 93%.

 Reflect and discuss (F)

An instructor asked her students how much time (to the nearest hour) they spent studying for the midterm. The data are displayed in the following histogram:



 14. What do the numbers on the _____ axis represent?

- A. The values of the number of hours studied.
- B. The count of students falling in each of the intervals.

 15. What do the numbers on the _____ axis represent?

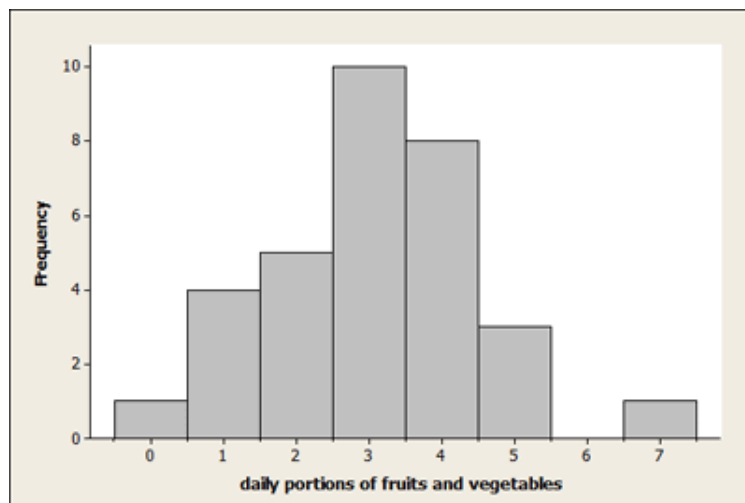
- A. The values of the number of hours studied.
- B. The count of students falling in each of the intervals.

16. What percentage of students study _____ or more hours for the midterm?

- A. 24%.
- B. 48%.
- C. 52%.
- D. 72%.
- E. 88%.

 Reflect and discuss (G)

Thirty-two students were asked the number of servings of fruits and vegetables they eat daily. The results are displayed in the histogram below.



17. How many of the students surveyed eat at least _____ servings of fruits and vegetables daily?

- A. 4.
- B. 8.
- C. 12.
- D. 20.
- E. 28.

18. What percentage of the students surveyed eat no more than _____ servings of fruits and vegetables daily?

- A. 20.
- B. 31.2%.
- C. 37.5%.
- D. 62.5%.
- E. 68.8%.

19. What proportion of the students surveyed eat exactly _____ servings of fruits and vegetables daily?

- A. 0.063.

- B. 0.094.
- C. 0.156.
- D. 2.
- E. 3.

Reflect and discuss (H)

A survey was conducted to see how many phone calls people made daily. The results are displayed in the table below:

Number of calls made	Frequency
1 - 4	16
5 - 8	11
9 - 12	5
13 - 16	3
17 - 20	1

 20. How many of the people surveyed make less than _____ phone calls daily?

- A. 3.
- B. 8.
- C. 9.
- D. 27.
- E. 32.

 21. How many people were _____?

- A. 20.
- B. 7.2.
- C. 36.
- D. 226.

Module 4 | Histogram (3 of 3)

Reflect and discuss (I)

 22. What is the most likely shape of the distribution of age of death from trauma (accident, murder, suicide, drug overdose, etc.) when represented by a histogram?

Recall that we talked earlier about the shape of the distribution of age of death from natural causes (heart disease, cancer, etc.). Use a similar type of reasoning for the age of death from trauma.

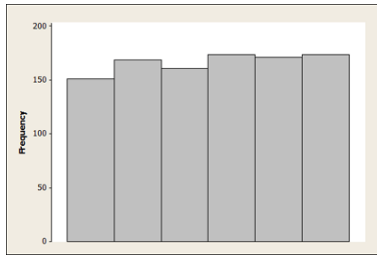
- A. Symmetric – uniform.
- B. Skewed left.
- C. Skewed right.
- D. Symmetric – unimodal.
- E. Symmetric – bimodal.

 23. What is the most likely shape of the distribution of the age at which a child takes its first _____?

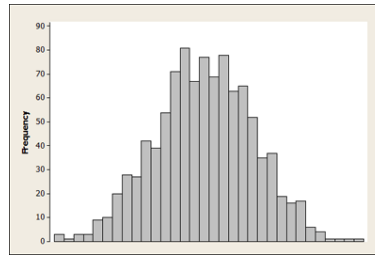
- A. Symmetric – uniform.
- B. Skewed left.
- C. Skewed right.
- D. Symmetric – unimodal.
- E. Symmetric – bimodal.

Reflect and discuss (J)

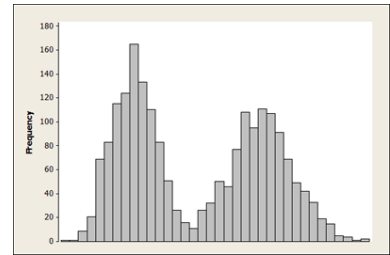
Match each histogram with the description of data that it would be most likely to represent.



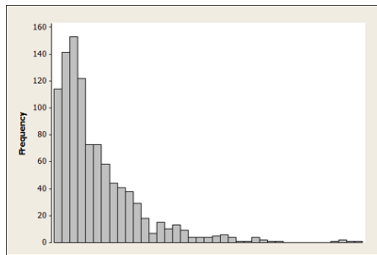
A.



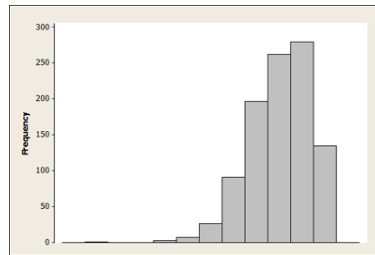
B.



C.



D.



E.

1. SAT Math scores of 1,000 future engineers and scientists.
2. Results of rolling a six-sided die 1,000 times.
3. Cholesterol levels of 1,000 adults.
4. Shoe sizes of 1,000 men and women.
5. Prices of 1,000 California homes.

Module 4 | Measures of Center (1 of 2)

Reflect and discuss (K)

Here are the number of hours that 9 students spend on the computer on a typical day:

1 6 7 5 5 8 11 12 15

 24. The _____ number of hours spent on the computer is {5, 5.5, 7.5, 8 }.

 25. The data from the above _____ students forms a { unimodal, bimodal, multimodal } distribution.

Reflect and discuss (L)

 26. Here are the number of hours that 9 students spend on the computer on a typical day:

1 6 7 5 5 8 11 12 15

The _____ number of hours spent on the computer is: { 7.00, 7.78, 8.23, 9.14 }

 Reflect and discuss (M)

 27. A recent survey asked 90 students, How many hours do you spend on the computer in a typical day?

Of the 90 respondents, 3 said 1 hour, 5 said 2 hours, 15 said 3 hours, 25 said 4 hours, 20 said 5 hours, 15 said 6 hours, 5 said 7 hours, 1 said 8 hours, and 1 said 9 hours.

What is the _____ number of hours spent on the computer?

- A. 0.5.
- B. 4.
- C. 4.44.
- D. 5.

 Reflect and discuss (N)

 28. Here are the number of hours that 9 students spend on the computer on a typical day:

1 6 7 5 8 5 11 12 15

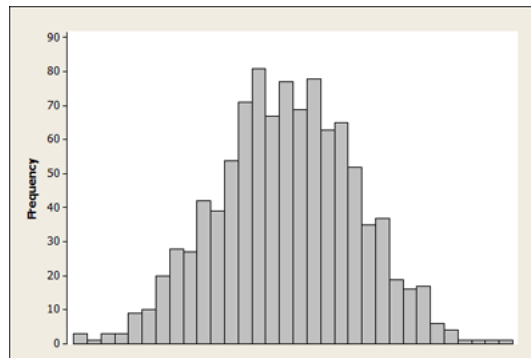
What is the _____ number of hours spent on the computer?

- A. 5.
- B. 6.5.
- C. 7.
- D. 7.5.
- E. 8.

Module 4 | Measures of Center (2 of 2)

 Reflect and discuss (O)

 29. In the histogram below, what will be the relationship between the mean and median of the collected data?



- A. The mean will be bigger than the median.
- B. The mean will be smaller than the median.
- C. The mean and median will be about the same.

 30. The Current Population Survey conducted by the Census Bureau records the incomes of a large sample of U.S. households each month. What will be the relationship between the mean and median of the collected data?

- A. The mean will be bigger than the median.
- B. The mean will be smaller than the median.
- C. The mean and the median will be about the same.

31. The SAT Math scores of 1,000 future engineers and physicists are recorded. What will be the relationship between the mean and median of the collected data?

- A. The mean will be bigger than the median.
- B. The mean will be smaller than the median.
- C. The mean and median will be about the same.

Reflect and discuss (P) Readability of Cancer Pamphlets

Background A study was done in order to find out whether pamphlets containing information for cancer patients are written at a level that the cancer patients can understand. Tests were administered to measure the reading levels of 63 cancer patients, and the readability levels of 30 cancer pamphlets were evaluated based on such factors as the lengths of the sentences and the number of polysyllabic words. Both the reading and readability levels correspond to grade levels, but patients' reading levels of less than grade 3 and above grade 12 cannot be determined exactly.

(Source: Short, Moriarty, and Cooley. (1995). "Readability of Educational Materials for Cancer Patients." *Journal of Statistics Education*, v.3, n.2)

The following tables indicate the number of patients at each reading level and the number of pamphlets at each readability level.

Patients' Reading Level	<3	3	4	5	6	7	8	9	10	11	12	>12
Count	6	4	4	3	3	2	6	5	4	7	2	17

Pamphlets' Readability Level	6	7	8	9	10	11	12	13	14	15	16
Count	3	3	8	4	1	1	4	2	1	2	1

Comment For both the reading level and readability level, the data are presented in a grouped form where the count represents the frequency of occurrence of that level. In the readability data, for example, the count of level 6 is 3 which means that the first three data points are 6 6 6; the count of level 7 is 3 which means that the next three data points are 7 7 7; the count of level 8 is 8 which means that the next eight data points are 8 8 8 8 8 8 8 8, and so on.

Answer the following questions:

32. The _____ of the readability level for the pamphlets is { 8, 16, 11, 7 }
Note that the frequencies of each level have already been determined.

33. Explain why you cannot calculate the _____ reading level of patients given the above data.

34. Find the _____ reading level of the patients:

Note that the data are already ordered – that's good!

We have $n = \{ 12, 30, 63, 93 \}$ observations of the reading level of cancer patients.

Since n is { even, odd }, the median reading level will be the { center observation, average of the two center observations } in the ordered data, which in this case is the { 31st, 32nd, 63rd } ranked observation.

The median reading level is therefore { 7, 7.5, 8, 9 }.

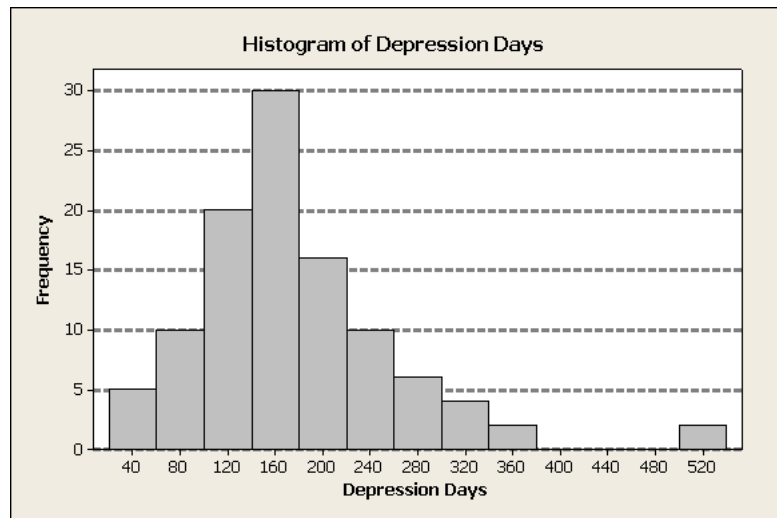
35. Find the _____ readability level of the pamphlets.

36. Can you conclude that the pamphlets are _____ to the patients' reading levels? Look carefully at the data.

 Reflect and discuss (Q) Measures of Center: Additional Questions

In the workplace, depression is a leading cause of absenteeism and loss of productivity (Greenberg, et al. 1993). To assess the degree to which people suffer from depression, prior to receiving treatment, data were collected on the number of days that 105 patients were depressed prior to starting a new treatment. These data are displayed in the following table and histogram:

Days	Count
[20–60]	5
[60–100]	10
[100–140]	20
[140–180]	30
[180–220]	16
[220–260]	10
[260–300]	6
[300–340]	4
[340–380]	2
[380–420]	0
[420–460]	0
[460–500]	0
[500–540]	2



 37. Which of the following is a possible value of the _____ number of days that patients were depressed?

- A. 53.
- B. 170.
- C. 220.
- D. 290.

 38. Using this same histogram of 105 patients, which of the following is most likely to be _____?

- A The mean will be larger than the median.
- B The median will be larger than the mean.
- C The mean and median will be about the same.

 39. Using this same histogram of 105 patients, what percentage of patients had _____ or more days of depression?

- A. 13.
- B. 23.
- C. 24.